

Apply filters to SQL queries

Project description

I am a security professional at a large organization. Part of your job is to investigate security issues to help keep the system secure. I recently discovered some potential security issues that involve login attempts and employee machines.

My task is to examine the organization's data in their employees and log_in_attempts tables. You'll need to use SQL filters to retrieve records from different datasets and investigate the potential security issues.

Retrieve after hours failed login attempts

To look into possible security incidents that happened after business hours, I'll query the **log_in_attempts** table using SQL filters. Specifically, I'll retrieve all records of failed login attempts that occurred after 18:00.

SQL Query:

```
SELECT *  
FROM log_in_attempts  
WHERE success = 0 AND TIME(login_time) > '18:00:00';
```

Retrieve login attempts on specific dates

To investigate this event, I will have to review all login attempts which occurred on this day and the day before. Use filters in SQL to create a query that identifies all login attempts that occurred on 2022-05-09 or 2022-05-08.

SQL Query:

```
SELECT *  
FROM log_in_attempts  
WHERE login_date IN ('2022-05-08', '2022-05-09');
```

Retrieve login attempts outside of Mexico

To examine suspicious activity with login attempts that happened outside of Mexico, I'll use SQL filters to get records from the **log_in_attempts** table where the country isn't Mexico (this includes both "MEX" and "MEXICO").

SQL Query:

```
SELECT *  
FROM log_in_attempts  
WHERE country NOT LIKE 'MEX%';
```

Retrieve employees in Marketing

To get information on these employee machines in the Marketing department I will need to query the employees table and utilize SQL filters.

SQL Query:

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Retrieve employees in Finance or Sales

To get information on these employee machines in the Finance or Sales department I will need to query the employees table and utilize SQL filters.

SQL Query:

```
SELECT *  
FROM employees  
WHERE department IN ('Finance', 'Sales');
```

Retrieve all employees not in IT

To get all employees who are not in the IT department, I will query the "employees" table using SQL filters.

SQL Query:

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

Summary

Through SQL queries, I've looked into possible security issues like after-hours login attempts, suspicious logins on certain dates, login activity from outside Mexico, and employee details by department and location. These queries help improve the organization's security by finding and addressing potential risks and weaknesses.